DECIPHERING THE ANTI-OBESITY MECHANISM OF *ACALYPHA INDICA* THROUGH INTEGRATIVE NETWORK PHARMACOLOGY AND MOLECULAR DOCKING

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1 OBJECTIVE

1 OBESITY IS NOT DRIVEN SOLELY BY EXCESS FAT ACCUMULATION

In many individuals, persistent overeating, altered food reward and impaired appetite control contribute to a sustained positive energy balance.

2 REGULATED BY CENTRAL NEURAL CIRCUITS

These behaviours are regulated by central neural circuits, particularly serotonergic and dopaminergic pathways that influence hunger, satiety, food reward and energy homeostasis.

3 *ACALYPHA INDICA*

A traditionally used medicinal plant with reported metabolic relevance, offers a potential source of bioactive phytochemicals.

4 KEY QUESTION

However, whether its metabolites can converge on molecular targets involved in central appetite and energy regulation **remains unclear.**

5 STUDY OBJECTIVE

This study therefore investigates the potential link between *A. indica* phytochemicals, obesity-associated molecular evidence and the neuro-energetic targets **HTR2A** and **SLC18A2.**

RESEARCH QUESTION

Can *Acalypha indica* metabolites target molecular pathways involved in eating behaviour and obesity?

WHY THIS STUDY ?

Understanding these molecular interactions may provide new insights into the potential of plant-based therapeutic strategies for obesity.

2 MATERIALS & METHODS

INTEGRATIVE COMPUTATIONAL WORKFLOW – DATA EVIDENCE CHAIN

Stage	Platform / Database	Input / Selection Criteria	Direct Output / Yield
1	IMPPAT Database	*Acalypha indica* constituent profile	Bioactive small-molecule library
2	Swiss TargetPrediction	Probability-based target scoring	Predicted human protein target list
3	Open Targets Platform	Obesity-associated disease signatures	Clinical druggable target dataset
4	NCBI GEO Datasets	GSE234889, GSE306336, GSE306884, GSE314750, GSE317489, GSE337548	Upregulated gene signatures in obesity
5	Cross-Intersection Analysis	Tripartite overlap matching	9 Common Therapeutic Targets
6	Pharmacophore Mapping	Ligand–target interaction feature mapping	4 Key Prioritized Phytochemicals
7	Molecular Docking & Scoring	Target Structures: HTR2A (7VOE) & SLC18A2 (8T69)	Binding score matrix (ΔG in kcal/mol)

TARGET RATIONALE

1 FROM 9 COMMON TARGETS TO 2 SELECTED TARGETS

9 COMMON TARGETS

- PTGER2 (Prostaglandin E Receptor 2)
- GABRA3 ($GABA_A$ Receptor α3)
- HTR2A ($5\text{-}HT_{2A}$ Receptor 2A)
- HTR2B ($5\text{-}HT_{2B}$ Receptor 2B)
- CCR5 (C-C Motif Chemokine Receptor 5)
- SSTR2 (Somatostatin Receptor 2)
- PTGER3 (Prostaglandin E Receptor 3)
- ADORA2B (Adenosine A_{2B} Receptor)
- SLC18A2 (Vesicular Monoamine Transporter 2)

BIOLOGICAL RELEVANCE & OBESITY ASSOCIATION
Targets are implicated in appetite regulation, reward pathways and neuro-energetic balance.

2 SELECTED TARGETS
HTR2A + SLC18A2
2 PRIORITIZED NEURO-ENERGETIC TARGETS

Obesity is not driven solely by peripheral fat accumulation; persistent overeating arises from altered brain circuits controlling appetite, food reward and energy balance.

HTR2A 7VOE — SLC18A2 (VMAT2) 8T69

Linked to serotonergic and monoaminergic pathways

Feeding behaviour | Reward | Energy balance

EVIDENCE & JUSTIFICATION

INHIBITION ADVANTAGE
- Inhibitory targeting is generally easier to investigate.
- Provides a direct and effective pharmacological strategy.
- Enhances translational potential for anti-obesity therapy.

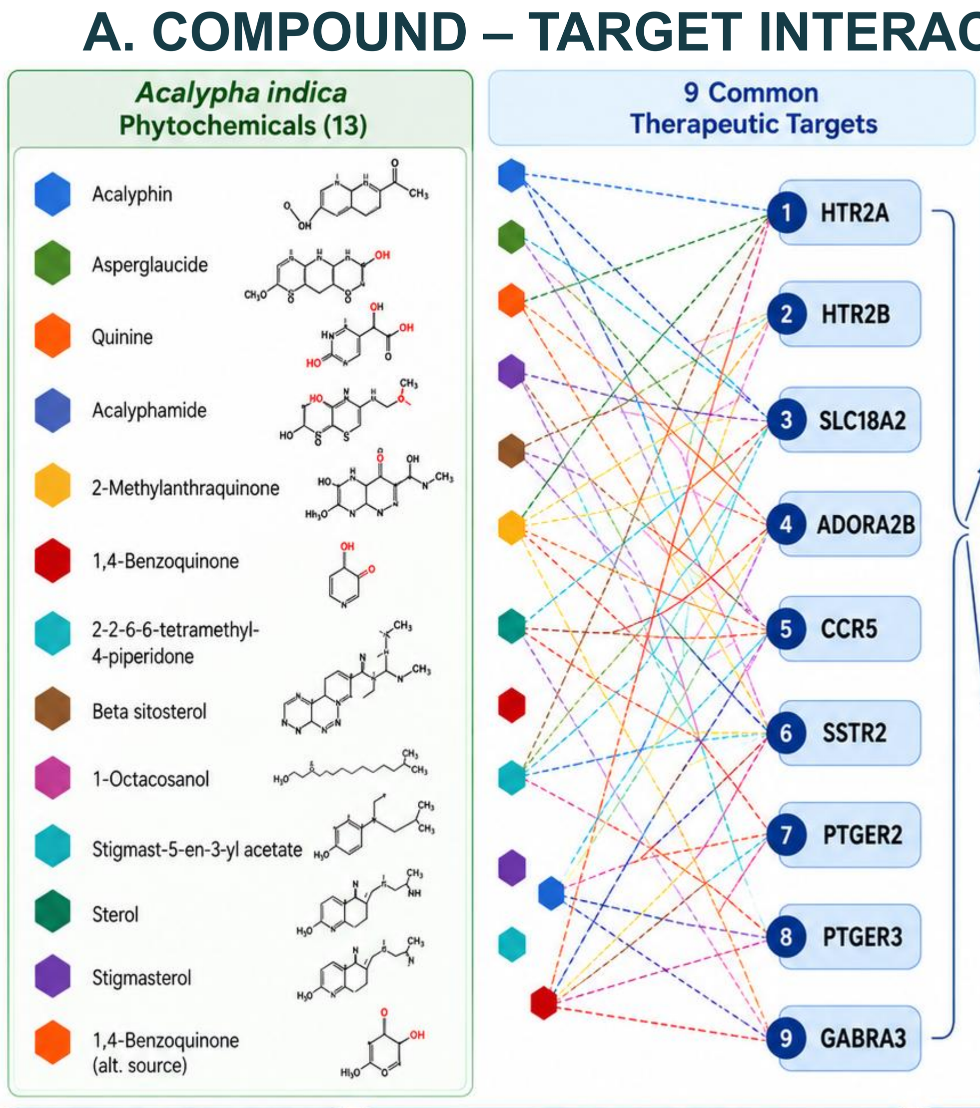
BIOLOGICAL RELEVANCE
- Both targets are central to obesity-linked neuro-energetic pathways.
- Regulate appetite, reward processing and energy balance.
- Their modulation can normalize maladaptive eating behaviour.

CLINICAL EVIDENCE
- Strong pharmacological agents exist for these targets.
- Therapeutic potential established in metabolic and neuropsychiatric disorders.
- Prior studies support their relevance in obesity and eating behaviour.

CLINICAL VALIDATION
- Ongoing and completed clinical studies support their relevance.
- Viable therapeutic targets for obesity intervention.
- Their simultaneous modulation may offer synergistic benefits.

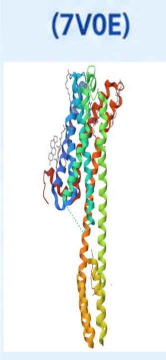
3 RESULTS

A. COMPOUND – TARGET INTERACTION NETWORK



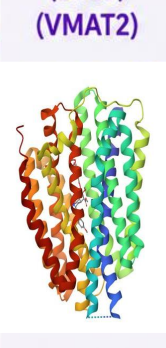
KEY NEURO-ENERGETIC TARGETS

HTR2A (7V0E)



- Serotonin 5-HT2A receptor (GPCR) widely expressed in the CNS.
- Regulates appetite, satiety signals and food intake.
- Modulates reward processing, impulsivity and mood.
- Influences energy balance through hypothalamic and corticolimbic circuits.
- Involved in hedonic feeding and emotional eating.
- Pharmacologically validated target for anti-obesity and appetite control strategies.

SLC18A2 (8T69) (VMAT2)



- Vesicular Monoamine Transporter 2 (VMAT2) located on synaptic vesicles.
- Responsible for vesicular storage of monoamines: dopamine, serotonin, norepinephrine.
- Regulates neurotransmitter availability and synaptic release.
- Critically involved in reward pathways and food motivation.
- Modulates feeding behaviour and reinforcement learning.
- Established target in eating disorders and obesity therapeutics.
- Inhibition/Modulation can reduce reward-driven overeating and

SCAN TO EXPLORE THE DATA

NETWORK STATISTICS

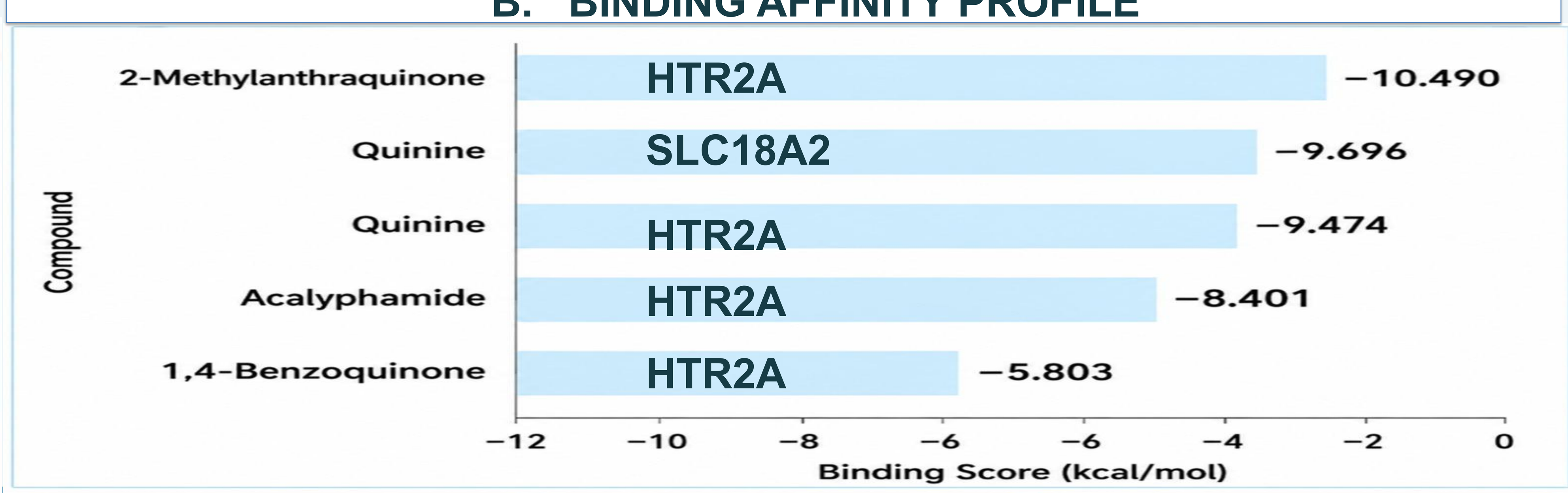
Total Phytochemicals:	13	Average Degree :	4.67
Total Targets :	9	Network Density :	0.583
Total Interactions :	42	Network Type :	Compound-Target

KEY INSIGHTS

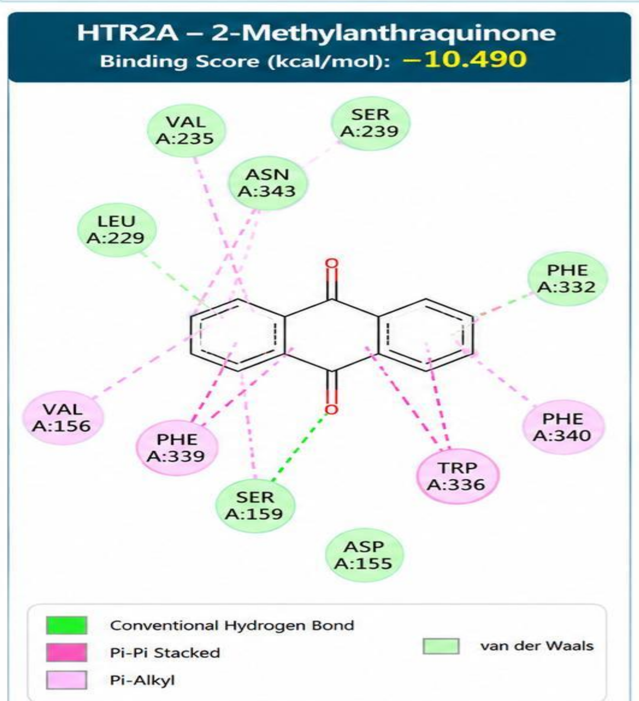
- HTR2A and SLC18A2 are central neuro-energetic targets linked to appetite regulation, reward pathways and energy balance.
- The network highlights multi-target potential of *Acalypha indica* phytochemicals in obesity and eating behaviour modulation.
- Polypharmacological interactions suggest synergistic effects through complementary neurotransmitter and GPCR-mediated mechanisms.

Note: The network illustrates predicted interactions between *Acalypha indica* phytochemicals and obesity/eating behaviour-related targets obtained through integrative computational analysis.

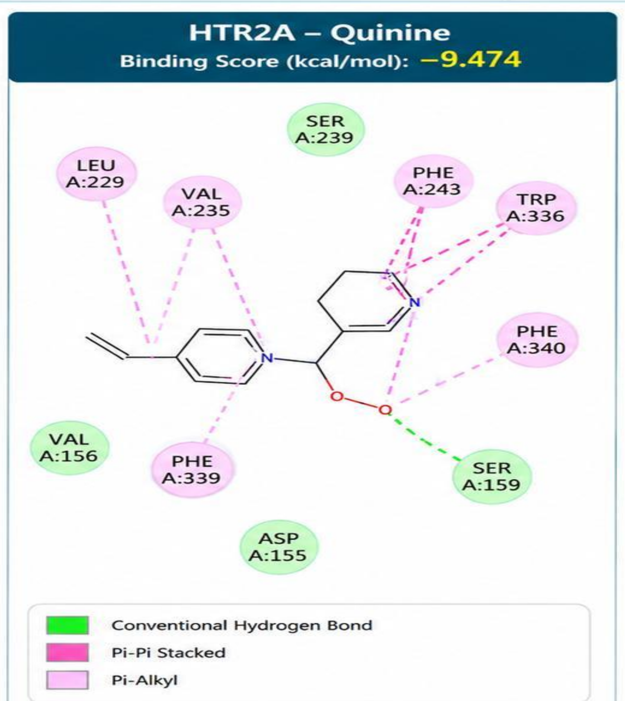
B. BINDING AFFINITY PROFILE



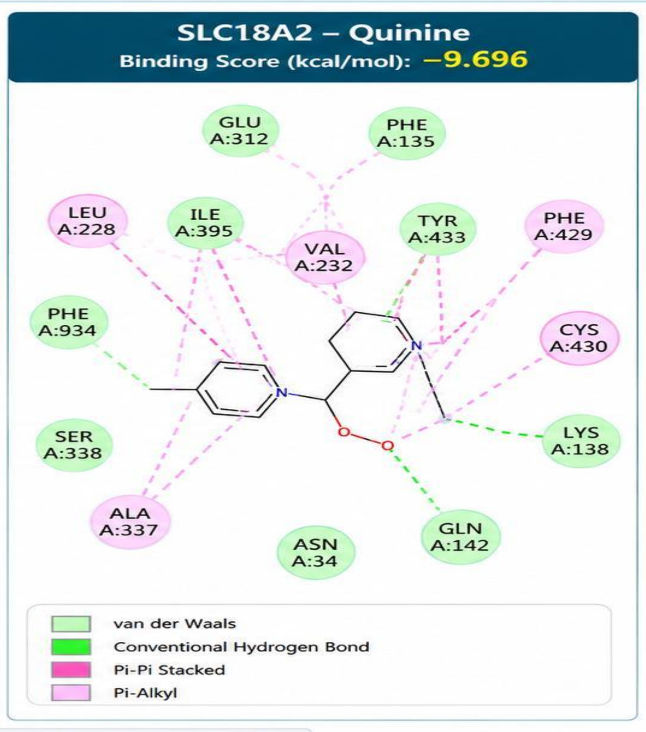
HTR2A – 2-Methylanthraquinone
Binding Score (kcal/mol): −10.490



HTR2A – Quinine
Binding Score (kcal/mol): −9.474



SLC18A2 – Quinine
Binding Score (kcal/mol): −9.696

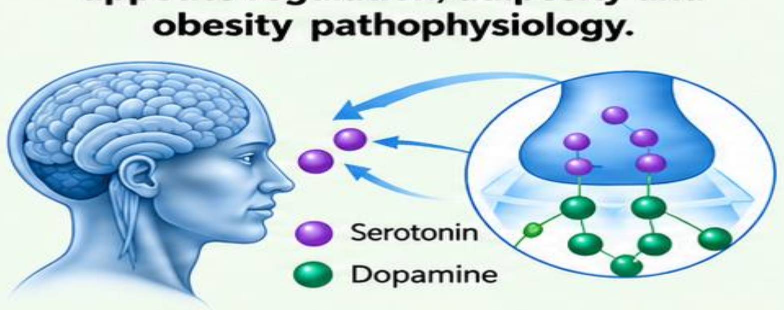


Note: Binding scores (kcal/mol) are obtained from molecular docking analysis. More negative scores indicate stronger binding affinity.

KEY OUTCOMES & FUTURE DIRECTIONS

BIOLOGICAL SIGNIFICANCE

HTR2A and SLC18A2 (VMAT2) connect central neurotransmitter signalling with pathways involved in feeding behaviour, reward processing and energy balance. Their modulation is strongly linked to appetite regulation, adiposity and obesity pathophysiology.



KEY MOLECULAR FINDINGS

1. **2-Methylanthraquinone → HTR2A**
   −10.490 kcal/mol
   (Strongest Binding)
2. **Quinine → SLC18A2 (VMAT2)**
   −9.696 kcal/mol
   (Very Strong Binding)
3. **Quinine → HTR2A**
   −9.474 kcal/mol
   (Very Strong Binding)

PROPOSED ANTI-OBESITY MECHANISM

Acalypha indica Phytochemicals
↓
HTR2A & SLC18A2 (VMAT2)
↓
Serotonergic & Monoaminergic Signalling
↓
Feeding Behaviour | Reward Processing
↓
Energy Balance
↓
Potential Influence on Obesity

NEXT STEP: EXPERIMENTAL VALIDATION

In vitro **target validation**
Biochemical assays, cell-based studies
↓
Pathway & mechanism analysis
Serotonergic / monoaminergic pathway assessment
↓
In vivo **obesity models**
Diet-induced obesity / genetic models
↓
Outcome evaluation
Food intake · Body weight · Adiposity
Energy expenditure · Metabolic parameters

TAKE-HOME MESSAGE

Our integrated computational analysis identifies **HTR2A** and **SLC18A2** as key neuro-energetic targets of *Acalypha indica*. Strong predicted interactions, particularly **2-Methylanthraquinone—HTR2A (−10.490 kcal/mol)** and **Quinine—SLC18A2 (−9.696 kcal/mol)**, provide a testable molecular basis for investigating the plant's potential role in regulating feeding behaviour and energy balance.

4 CONCLUSION

- This study provides an integrated computational framework linking *Acalypha indica* phytochemicals with obesity-associated neuro-energetic mechanisms.
- By converging phytochemical, target-prediction, transcriptomic and molecular docking evidence, HTR2A and SLC18A2 (VMAT2) emerged as the most relevant targets governing serotonergic and monoaminergic regulation of appetite, food reward and energy balance.
- The strong predicted binding of 2-Methylanthraquinone and Quinine further supports a plausible molecular basis through which *Acalypha indica* may modulate these pathways.
- Collectively, these findings position A. indica as a promising source of anti-obesity lead compounds and establish a strong, testable foundation for experimental validation of its therapeutic potential.

Scan for References